

Alliance Manchester Business School

Information Technology Management for Business with Industrial Experience

# Design, Deployment, and Sustainability: A Grounded Theory Analysis of Energy Demand in Open- and Closed-Source Large Language Models

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Project Submission Date: 06.05.2025

Word Count: 8,741

## Abstract

The rapid development and deployment of large language models (LLMs) have prompted increasing scrutiny of their environmental footprint, particularly in relation to energy consumption. This study investigates how the design and deployment models of open- and closed-source LLMs shape their energy demand, environmental impact, and capacity for mitigation. Employing a constructivist grounded theory methodology, this research systematically analyses a curated corpus of academic literature and empirical sources to develop an explanatory framework. The study identifies and compares the architectural, operational, and infrastructural differences between open-source models (such as LLaMA and DeepSeek) and closed-source models (such as ChatGPT and Claude). While open-source models promote distributed deployment and often foster innovative, resource-efficient designs, they can inadvertently lead to redundant computational efforts across independent users. In contrast, closed-source models centralise computation in large-scale data centres, offering optimised hardware utilisation at the cost of opacity and limited accountability.

Using open, axial, and selective coding techniques, the analysis reveals four core dimensions underpinning energy use in LLMs: drivers of energy demand, mitigation strategies, outcomes and governance mechanisms. The findings demonstrate that factors such as model size, GPU usage, inference strategies, and deployment practices critically influence energy profiles. Moreover, the study shows that sustainable AI deployment is a sociotechnical challenge requiring not only technical interventions but also institutional governance and policy frameworks. The research contributes a conceptual framework that maps the trade-offs inherent in different LLM ecosystems and highlights the role of transparency, standardisation, and shared ownership in enabling energy-conscious AI. This study offers actionable insights for developers, policymakers, and platform architects striving to balance innovation with environmental stewardship in the age of large-scale AI.

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# 1. Introduction

## 1.1 Background and Context

Artificial intelligence has experienced a rapid expansion in recent years, epitomised by the surge of large language models (LLMs) and other foundation models. In 2023, developers launched 149 foundation models, more than twice the 2022 total, with 65.7 % released as open-source models, compared with 44.4 % in 2022 and just 33.3 % in 2021 (Maslej, et al., 2024). This trend reflects a growing shift towards open-access AI development. LLMs such as OpenAI's GPT series, Anthropic's Claude, Meta's LLaMA and various community-developed models have quickly become central to modern applications, powering conversational agents, search assistants, code generation tools and more. Advances in model training techniques, particularly reinforcement learning from human feedback (RLHF), have improved the alignment of LLM outputs with human preferences, expanding their usability across domains like medicine, education and software development. As a result, systems like ChatGPT have rapidly permeated everyday life, demonstrating the transformative potential of LLM technology (Teubner, et al., 2023).

Despite their impressive capabilities and broad adoption, LLMs also introduce significant challenges. One key concern is the enormous computational resources required to train and operate these models (Jiang, et al., 2024). Frontier models now demand unprecedented amounts of compute power (Maslej, et al., 2024). LLMs can perform a wide array of natural language processing tasks (from summarisation and translation to complex reasoning) without task-specific fine-tuning, but this generalized ability comes at the cost of high model complexity and computational load. These technical and ethical challenges further heighten the need for careful deployment and management of LLMs (Strubell, et al., 2019).

In particular, the deployment model, whether an LLM is openly released for distributed use or kept closed source in centralised datacentres, may shape how and where energy is consumed. Open-source LLMs are often released along with their model weights, allowing anyone to run them on local or custom hardware. This can decentralise the compute load, potentially reducing reliance on massive data centres but also possibly leading to duplicate efforts. By contrast, closed-source LLMs (like many commercial offerings) are typically hosted by providers in large-scale cloud infrastructures, which concentrate energy use in big

server farms. These differing approaches bring trade-offs in terms of energy efficiency, transparency, and applicability of mitigation strategies.

## 1.2 Problem Statement

LLMs promise transformative benefits across industries, but their burgeoning energy requirements pose a pressing sustainability challenge. The problem at the heart of this research is the insufficient understanding of how the design and deployment model of LLMs (open-source versus closed-source) affects their energy consumption and environmental impact. While recent work has highlighted the general environmental cost of large AI models (Schwartz, et al., 2019; Ren, et al., 2024) there is a notable gap in comparative knowledge regarding different development approaches. Open-source and closed-source LLMs may follow distinct paths in model design, infrastructure, and usage patterns, which could lead to divergent energy profiles and carbon footprints. However, the existing literature offers limited insight into these differences, partly because closed-source providers often do not disclose detailed resource usage data (Kneese & Young, 2024). This opacity makes it difficult for the research community to gauge how, for example, a centrally hosted proprietary model's efficiency stacks up against that of a community-developed open model running on distributed hardware.

Addressing this problem is crucial for both the AI field and wider society. Without a nuanced understanding of how design and deployment choices influence energy demand, efforts to make AI sustainable might be misdirected or insufficient. Organisations and policymakers increasingly recognise the importance of curbing the environmental footprint of AI; for instance, recent policy proposals in the EU mandate transparency in AI energy use and efficiency reporting (European Parliament, 2023). Yet, in the absence of comparative analysis, crafting guidelines or best practices for sustainable LLM deployment is challenging. This research responds to the identified gap by examining the interplay between LLM design/deployment models and their energy and environmental implications. The aim is to develop a grounded understanding of the trade-offs involved and to identify opportunities for mitigation.

To achieve this, the research adopts an exploratory qualitative approach, using grounded theory to systematically derive patterns and theories from empirical data on LLM usage and

design characteristics. A grounded theory methodology is well-suited here because the topic is relatively growing and under-theorised; it allows theory to emerge from data rather than imposing existing frameworks (Charmaz, 2014).

### 1.3 Research Question

Considering the above context and problem statement, this study is guided by the following primary research question:

*How does the design and deployment model of open and closed-source LLMs shape their energy demand, environmental consequences, and mitigation capacity?*

This question will direct the research toward comparing open-source versus closed-source LLM approaches in terms of energy efficiency and sustainability. It encapsulates the key focus areas: understanding the energy demand associated with each model type, examining the environmental consequences (such as carbon footprint and resource usage), and exploring the mitigation capacity, meaning the ability of each approach to minimise or offset those impacts.

## 2. Literature Review

### 2.1 Introduction to Literature Review

The aim of this chapter is to critically review the existing academic literature relevant to the research question, which investigates the trade-offs in security and energy-efficiency for open-source versus closed-source LLMs. Given the rapid expansion of LLM technologies and their growing societal and environmental implications, it is necessary to explore the two domains: the energy-efficiency risks associated with LLM deployment, and the environmental cost arising from the increased energy-demand with the wide use of LLMs. Furthermore, the chapter considers how issues of transparency, accountability, and reporting standards differ across open-source and closed-source models as this is a major limitation in the availability of data and reports on the issues.

The review adopts a structured approach, first defining the key technical concepts surrounding LLMs, before examining thematic strands of research related to energy efficiency. Finally, the chapter identifies gaps in the existing body of knowledge, this providing the rationale for the current study and informing the development of the grounded theory framework which will be applied later in this research.

### 2.2 Large Language Models (LLMs): An Overview

LLMs have emerged as a transformative force in artificial intelligence, significantly reshaping interactions between humans and machines. LLMs are powerful neural architectures trained on extensive datasets, enabling them to understand, generate, and manipulate human language with proficiency that has never been seen before (Kamath, et al., 2024).

LLMs, such as OpenAI's GPT series, DeepSeek, and Meta's LLaMA, operate primarily on the transformer architecture, characterised by self-attention mechanisms that efficiently capture dependencies in sequential data without relying on recurrence (Liu, et al., 2025).

LLMs typically rely on an autoregressive transformer architecture, where each word or token is sequentially generated based on an initial prompt and all previously produced tokens (Vaswani, et al., 2017). Transformers leverage attention mechanisms to selectively focus on

relevant input information, enhancing their ability to process contextually rich and extensive textual data, thus allowing models like GPT-4 to reach up to hundreds of billions of parameters (Kumar, 2024).

The historical evolution of LLMs has seen a progression from rule-based and statistical language models to advanced neural architectures that employ large-scale training on diverse and vast textual datasets (Kumar, 2024). Modern LLMs incorporate training methods such as reinforcement learning from human feedback (RLHF) to align model outputs with human preferences, thereby improving the usability and appropriateness of responses (Teubner, et al., 2023). This advancement has significantly expanded their application scope beyond simple, traditional text generation, reaching into fields like medicine, education, and programming. LLMs such as ChatGPT have also become a major part of many people's daily lives.

Despite their impressive capabilities, LLMs present notable challenges, including computational demands, interpretability, and risks of producing biased or "hallucinated" outputs that appears plausible but is inaccurate or contextually inappropriate (Kumar, 2024) (Teubner, et al., 2023).

Furthermore, LLMs' operational costs and energy consumption have raised significant environmental concerns, emphasising the need for sustainable and energy-efficient AI solutions (Jiang, et al., 2024). As such, environmental impact and model safety have become critical factors in evaluating the trade-offs between different LLM deployment strategies, particularly in the context of open-source versus closed-source models, where transparency, accountability, and reporting standards differ substantially (Teubner, et al., 2023).

## 2.3 Energy Efficiency and Environmental Impact of LLMs

### 2.3.1 Green AI and the Environmental Cost of ML

The rapid growth of deep learning has led to exponentially increasing computational demands and subsequently environmental costs. Schwartz et al. (2020) introduced the concept of Green AI, describing it as artificial intelligence research aimed at achieving reliable outcomes without increasing, and ideally decreasing, the computational resources required. This

concept arose in contrast to “Red AI,” the dominant trend of pursuing state-of-the-art results at any expense of compute (Schwartz, et al., 2019). Such Red AI advances have come with steep energy requirements and a surprisingly large carbon footprint, ironically at odds with the brain’s efficiency that inspired neural networks. Green AI advocates for the incorporation of more sustainable practices during the development, deployment, inference, and fine-tuning of AI systems.

### 2.3.2 Factors Influencing the Energy Demand of LLMs

The energy demand of LLMs is influenced by multiple interrelated factors, spanning from model architecture and hardware efficiency to operational practices.

Firstly, model size significantly impacts energy consumption. Larger models, such as Meta’s Llama-3-70B, require more computational resources, thus consuming higher energy levels compared to lightweight models like Gemma-2B-it (Ren, et al., 2024). The volume of computations necessary for these large models leads to an extensive utilisation of computing resources, notably GPUs, which elevates their energy requirements. As model parameters increase, so do energy requirements, exacerbating the environmental impacts associated with their deployment (Argerich & Patiño-Martínez, 2024). This scaling relationship presents a critical consideration in the design and deployment strategies for LLMs, particularly in environmentally sensitive applications.

Secondly, the hardware and infrastructure efficiency play crucial roles. High-performance GPUs, essential for LLM inference, significantly influence the energy profile of model deployment (Jiang, et al., 2024). High-performance GPUs, which are essential for efficiently conducting LLM inference, represent one of the most significant factors in energy consumption. Advances in GPU technologies and architectures have continuously sought to balance performance and energy efficiency with increased focus on performance due to the high computational needs of these systems. Techniques such as PCCL (Power-aware Collective Communication Library) developed by researchers enable adjustments to GPU performance in real-time, reducing energy consumption by lowering operational frequencies during less computationally intensive tasks without compromising the quality of outputs (Jia, et al., 2024).

Operational factors such as batch size and inference strategies further affect energy usage significantly. Batch size, or the number of inference tasks processed simultaneously, is a critical operational parameter. Larger batch sizes typically enhance energy efficiency by maximizing GPU utilisation and improving throughput (Patel, et al., 2024). However, larger batch sizes can also introduce increased latency due to additional overhead from parallel processing and resource contention, potentially reducing overall responsiveness (Kakolyris, et al., 2025). This presents challenges during inference as requests may come at any time and they usually have vastly different input and output lengths. This is especially a problem for closed-source LLMs where the processing is completed in large data centres and the inputs are from a wide range of people, making it difficult to reach accurate predictions.

Finally, deployment practices and operational scaling significantly influence overall energy efficiency. The implementation of automated scaling mechanisms that dynamically allocate computational resources in accordance with fluctuating workloads helps avoid overprovisioning and underutilisation of computing resources (Patel, et al., 2024). These mechanisms ensure that computational resources, and thus energy consumption, match actual demand closely, reducing unnecessary energy usage during periods of low activity and scaling up appropriately during peak periods (Kakolyris, et al., 2025). Such adaptive strategies are crucial for maintaining operational efficiency and sustainability, particularly in cloud-based and large-scale deployments of LLMs.

Collectively, these factors highlight the complexity involved in managing energy demand in LLMs. They underscore the necessity of integrated approaches combining advanced hardware technologies, adaptive operational practices, and methodological innovations to achieve sustainable deployment of artificial intelligence technologies.

## 2.4 Open-Source versus Closed-Source LLMs

### 2.4.1 Open Source LLMs

#### *2.4.1.1 LLaMA*

In contrast to the above, Meta AI's LLaMA was conceived explicitly to maximize performance for a given compute budget. It is an open family of models (released to researchers and now partly open source) that range from 7B to 65B parameters (Touvron, et

al., 2023). Touvron et al. (2023) has observed that while training a large model to achieve a certain performance level might initially be cheaper, training a smaller model for a longer duration ultimately results in lower inference costs. This means that for many tasks one can use a 13B model (which easily fits on a single GPU or even a high-end laptop) instead of calling a 175B model that requires an entire server node. By right sizing the model, LLaMA dramatically lowers the computational requirements for inference. Meta initially shared the weights with academics in early 2023, and later versions, such as Llama 2, were released under a permissive licence. This open access allows others to build upon the existing model rather than retraining entirely new models, a practice which Meta explicitly suggests will help to reduce future carbon emissions as the training has already been completed (Touvron, et al., 2023).

Meta trained LLaMA using its in-house AI Research SuperCluster (RSC) infrastructure. The mention of 2,048 NVIDIA A100 80GB GPUs over 5 months gives a sense of the scale of hardware committed (Touvron, et al., 2023). The energy source for Meta's compute is relatively sustainable, Meta has a strong renewable energy procurement program for its data centres, aiming for net-zero emissions. Running AI on local devices can be a big energy win in aggregate: it saves the energy of data centre servers and network transfer and leverages the highly efficient AI accelerators in modern consumer hardware. It also means the energy use is distributed and the energy demand from running and cooling the data centres doesn't factor in (Luo, et al., 2013).

#### *2.4.1.2 DeepSeek*

DeepSeek is a newer entrant to the LLM market that embodies the open-source, efficiency-driven ethos in a slightly different way. Developed by a Chinese AI startup, DeepSeek's flagship model DeepSeek-V3 boasts an eye-popping 671 billion parameters, but crucially it uses a Mixture-of-Experts (MoE) architecture (DeepSeek, 2025). In an MoE model, the parameters are divided among many expert subnetworks, and for each given input token only a subset of those are utilised. In effect, its effective model size per inference is 37B, which makes it much more computationally efficient than a dense 671B model would be. This design lets DeepSeek achieve extremely high capacity (diverse knowledge and skills from many experts) while keeping the inference cost per token on par with a 30B model (Bi, et al., 2024). DeepSeek released their model openly (both DeepSeek-V3 and a reasoning-focused

DeepSeek-R1). The weights are free for anyone, and within weeks, over 700 derivative models based on DeepSeek appeared on Hugging Face, collectively downloaded millions of times (Smith, 2025).

## 2.4.2 Closed Source LLMs

### 2.4.2.1 ChatGPT

ChatGPT is built on OpenAI's GPT series of transformer models. GPT-3, for example, is a 175-billion-parameter autoregressive language model (Brown, et al., 2020). GPT-4 is more advanced than GPT-3 and the exact details of the model have not been released, although it is estimated that it has more than 1 trillion parameters (EPRI, 2024). These models are trained with reinforcement learning from human feedback to fine-tune responses, improving quality but adding extra training cycles. Such enormous model size and training requirements translate to very high computational demand and energy cost (Jiang, et al., 2024). Notably, training the GPT-3 model alone consumed an estimated 1,287 MWh of electricity, emitting approximately 552 tons CO<sub>2</sub> which roughly translates to the energy annual power usage of 120 U.S. homes (Zewe, 2025). According to Li, et al. (2025) Training the GPT-3 language model in Microsoft's advanced U.S. data centres can directly consume 700,000 litres of clean freshwater through evaporation, a fact that has remained undisclosed. A report by EPRI (2024) states that the energy consumption to train ChatGPT was around 62,000 MWh. This further amplifies the environmental impact of LLMs as the only concern regarding the training and inference of such large models is not the energy use but also the water used while cooling the data centre. The lack of transparency makes it difficult to assess improvements or hold providers accountable for sustainability.

Additionally, running ChatGPT at scale requires a massive cloud infrastructure. OpenAI's models are served via Microsoft Azure's supercomputing clusters. In fact, Microsoft built a dedicated AI supercomputer for OpenAI featuring 285,000 AMD CPU cores and 10,000 NVIDIA GPUs interconnected with high bandwidth networking (Cavanell, 2023). OpenAI's closed-source deployment means the model runs only on their servers, but this also allows centralised optimisations which will be discussed in more detail later.

#### *2.4.2.2 Claude*

Anthropic's Claude is another proprietary LLM service, developed by a company founded by former OpenAI researchers. Claude's architecture is similar in foundation, an autoregressive transformer, but Anthropic initially focused on making it smaller and safer (Anthropic, 2023).

Anthropic has invested heavily in AI supercomputing, in part through cloud partnerships. Initially, Anthropic utilised both Google Cloud and Amazon Web Services (AWS) for training Claude models (Anthropic, 2023). They announced a collaboration with AWS to create a new supercomputer of 'hundreds of thousands of AWS Trainium2 chips' for next-generation Claude training (Anthropic, 2024). This cluster is expected to provide five times more total compute power (in exaflops) than what Anthropic used to train Claude 2 (Anthropic, 2024). This is a clear indicator that the next Claude model will push even higher in model size or complexity and thus require more energy if run conventionally (Jiang, et al., 2024).

Unlike OpenAI, Anthropic has actively explored efficiency measures for inference. One such strategy is model distillation: Anthropic has shown that a smaller "Claude Instant" model can be derived from the large model, retaining much of its accuracy at a fraction of the size (Anthropic, 2024). Claude remains a large, closed model, running in Anthropic's cloud. Every user query to Claude involves data traveling to AWS data centres and being processed on power-hungry accelerators. The centralised approach does allow Anthropic to manage energy use in one place and potentially keep utilisation high (serving many users on shared hardware) but increases the environmental impact (Rillig, et al., 2023).

## 3. Methodology

### 3.1 Introduction to Methodology

This chapter outlines the research design and methodological steps taken to investigate how energy efficiency manifests in LLM research, and how this trade-off is prioritised in open-source versus closed-source models. The study adopts a qualitative, interpretivist paradigm and an inductive reasoning approach, meaning it seeks to build understanding and theory from observations rather than test a predetermined hypothesis. Grounded theory is a systematic research approach that guides the collection of data and offers clear strategies for their analysis. Its primary goal is to build theory that provides a conceptual understanding of key issues or phenomena within the context being studied (Charmaz & Thonberg, 2021). A constructivist grounded theory methodology was chosen as the framework for inquiry, given its strength in developing theory grounded in data and accommodating emerging concepts through iterative analysis (Charmaz, 2014).

This study is also inductive and exploratory in its approach. Rather than starting with a formal hypothesis about how energy efficiency is balanced in LLM projects, the research began with broad questions and allowed patterns, concepts, and eventually theoretical insights to emerge from the data itself. An inductive strategy is appropriate because the phenomena of LLM energy-efficiency and trade-offs (especially comparing open vs. closed models) are not yet well-theorised; thus, we aimed to build theory from the ground up. The grounded theory methodology naturally complements this inductive stance (Tie, et al., 2019). Throughout the process, the constant comparative method, a hallmark of grounded theory, was used, meaning data elements (the texts and codes) were continually compared with each other to refine emerging categories and interpretations.

### 3.2 Grounded Theory Research Design

#### 3.3.1 Rationale for Choosing Grounded Theory

Given the exploratory nature of the research question, grounded theory (GT) was selected as the methodological approach. GT is a qualitative research methodology designed to generate theory from data in a systematic yet flexible manner (Charmaz, 2006). It is particularly suitable when a phenomenon is not well researched, the goal is to develop a theory that

explains the underlying processes involved in the study.” (Tie, et al., 2019). In the context of this study, there is limited established theory specifically addressing how energy efficiency trade-offs are handled during LLM development and inference, especially comparing open-source and closed-source models. Grounded theory allows the researcher to delve into this relatively uncharted territory and let patterns emerge from the data without being constrained by an existing theoretical framework.

Another reason for choosing grounded theory is its ability to handle complex processes and trade-offs. The research question essentially asks about a process (how trade-offs manifest and are prioritised) that likely involves multiple factors and dimensions (environmental considerations, technical constraints, organisational and societal priorities, etc.). GT is well-suited to unravel such complexity by breaking down data into codes and categories and then building up an integrative theory or framework.

Lastly, adopting a grounded theory design aligns with recent innovative research in the information systems and AI domain. For instance, Sammon et al. (2024) used a grounded theorising approach enhanced with AI assistance to analyse a set of technology research abstracts, demonstrating that GT can be applied to literature data to generate insightful conceptual models. The methodology used was adapted from the one used by Turobov et al. (2024) but the custom GPT was only used for initial (open) coding of the documents.

### 3.3 Prompt Engineering and GPT Construction

To incorporate the "Chain of Thought" (CoT) and "Role Prompting" techniques in prompt engineering for ChatGPT, a structured sequence of logical steps was devised, closely emulating human reasoning processes to enhance semantic understanding and responses from the model (Marvin, et al., 2024). The CoT approach was specifically operationalised by explicitly instructing ChatGPT to undertake step-by-step reasoning when identifying, analysing, and generating codes for significant phrases relevant to energy efficiency. This deliberate guidance encouraged ChatGPT to methodically build upon each analytical step, ensuring outputs reflected comprehensive and insightful syntheses rather than surface-level interpretations (Lee, et al., 2024). Simultaneously, the "role prompting" technique was employed by designating the GPT explicitly as an objective, qualitative analysis researcher tasked with systematically coding textual data. Assigning this clearly defined role directed the

model's cognitive resources to emphasise domain-specific knowledge and maintain consistent objectivity, thus significantly improving the accuracy and contextual relevance of the generated codes (Gao, 2023). The integration of both CoT and role prompting techniques created a more effective structured reasoning environment that is critical for robust open coding practices in academic research. This combined methodological approach not only facilitated greater precision in code generation but also increased the depth of semantic analysis by enabling ChatGPT to emulate human-like interpretive and reflective capabilities (Henrickson & Meroño-Peñuela, 2023).

Additionally, a custom GPT called “Qualitative Edge” was designed to support this research. The GPT was provided with a set of instructions regarding its role, supplemented by three foundational documents on qualitative analysis. These documents included detailed explanations of qualitative analysis principles and provided explicit examples of open coding paired with researchers' thought processes, enhancing the GPT's ability to replicate detailed qualitative coding workflows (Lee, et al., 2024). The details for the custom GPT model and the instructions provided by the user can be found in 9.1 Appendix 1: Structure of the Custom GPT Model.

## 3.4 Data Collection

### 3.4.1 Data Sources and Sampling Criteria

Qualitative research utilises textual data as empirical material (Flick, 2007a). We can classify potential sources of data into two types: data that you need to produce and data that already exists (Rapley, 2018a). These are also known as primary and secondary data sources. This research utilises secondary sources of data, specifically journal articles. A total corpus of 45 abstracts was established for the purposes of this research and a total of 10 journal articles were chosen for analysis.

Purposive sampling guided the initial abstract selection, ensuring the inclusion of highly relevant and content-rich data. Abstracts had to focus explicitly on large language models, broadly encompassing energy-efficiency and environmental factors. Within this framework, priority was given to studies addressing energy efficiency and mitigation techniques and 10 journal articles were picked from this corpus for analysis based on their abstracts.

### 3.5 Data Analysis

Data analysis followed the classic grounded theory approach of open coding, axial coding, and selective coding (Corbin & Strauss, 2008). The constant comparative method was applied throughout these stages to continually refine the clusters and their relationships. Notably, the initial open coding was aided by a custom GPT model (similar to the research by Turobov, et al. (2024)) under human supervision (a human-in-the-loop approach), whereas axial and selective coding was performed manually by the researcher. Each stage of analysis is detailed below.

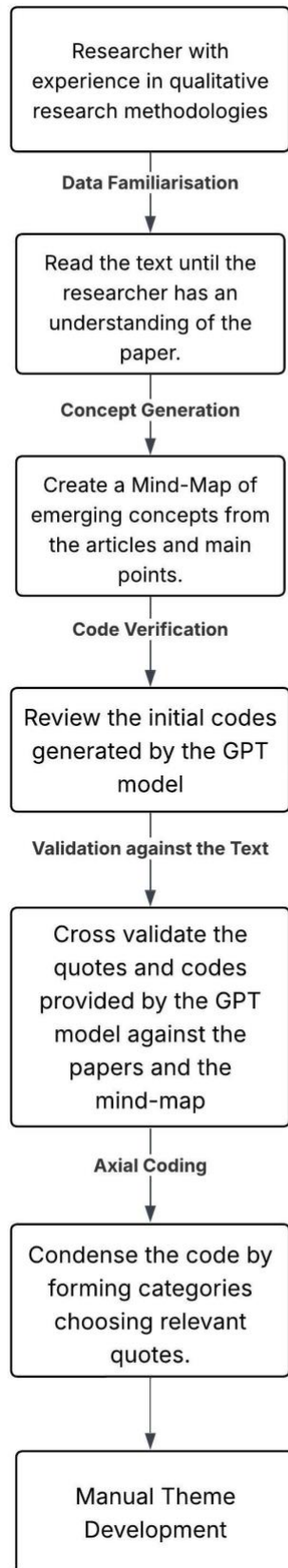
#### 3.5.1 Open Coding

Open coding was the first analytical step, where the data was broken into discrete concepts and each was assigned a code. Open coding helps a researcher identify the study's emerging direction, allowing them to refine their conceptual focus and concentrate on a specific social issue (Glaser, 2016). To expedite and enrich this stage, the researcher used a custom GPT Model, Qualitative Edge, as an assistant in a human-in-the-loop fashion. The researcher prompted the GPT with a set of instructions to analyse the paper uploaded and create open codes with direct references from the articles.

After open coding all 10 articles, the researcher organised related codes into a set of preliminary categories. For instance, several codes related to batch optimisation techniques were grouped under “Operational Efficiency & Energy-Sensitive Scheduling”.

QualitativeEdge helped highlight patterns and terminology in an unbiased fashion, but final coding decisions remained firmly with the human researcher, keeping the interpretation grounded in the data. All the quotes and the open codes were verified by the researcher after the analysis to ensure that the model did not suffer from hallucinations. These initial categories summarised the recurring topics and laid the groundwork for the next phase of analysis. The process followed for this can be seen in Figure 1. The GPT tended to shorten quotes, sometimes bringing them out of context, which enhances the need for the human-in-the-loop approach. The researcher tried mitigating this with updating the prompt used asking for direct quotes but this effort was not successful, therefore each code was manually verified against the article and the mind-map created by the researcher while they were reviewing the papers.

### Manual Thematic Analysis



### GPT Model Thematic Analysis

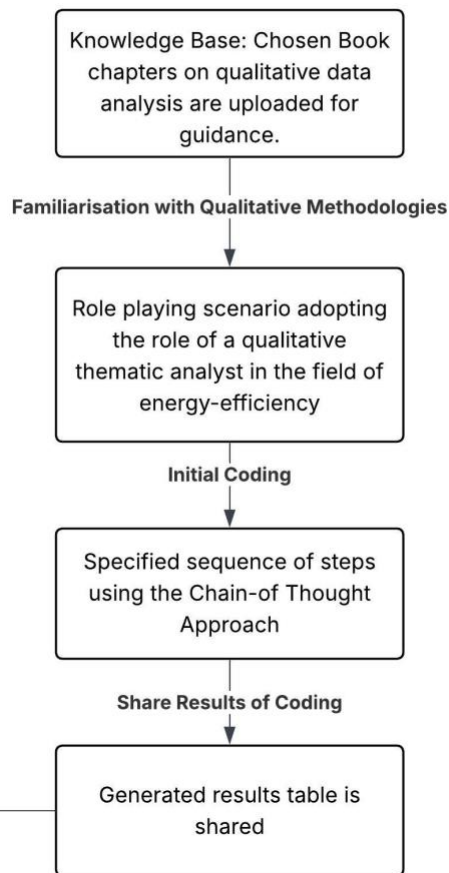


Figure 1: Open Coding Methodology using a GPT Model

### 3.5.2 Axial Coding

Axial coding was the next phase, focused on identifying and uncovering relationships between the emerging codes and the clusters identified. In this phase, the researcher sought to organise the categories more systematically by identifying how each related to the broader processes and phenomena. The clusters formed were collected under 4 broad categories: Drivers of Energy Demand, Mitigation Strategies, Outcomes & Oversight, Monitoring & Governance. Examples of open codes can be seen in Appendix 3: Sample Open Coding by the Custom GPT Model "QualitativeEdge".

In this domain, the core phenomenon identified was “LLM Energy Consumption & Environmental Impact” Categories were structured around this phenomenon according to their relationships with each other. Throughout axial coding, the constant comparative method was rigorously applied: each emergent category was repeatedly compared against the raw data and other categories to validate its relevance and clarify its role in the broader frameworks.

### 3.5.3 Selective Coding

Selective coding was the final stage of the analysis, aiming to integrate the separate axial coding frameworks into a coherent, higher-level theoretical structure showcasing the relationship between emerging clusters. In this phase, the researcher identified the core phenomena that captured the essence of each analytical domain. The purpose was to develop a conceptual framework that would later support a comparative evaluation of open-source and closed-source LLMs regarding energy-efficiency factors and mitigation techniques.

Through constant comparison, the researcher validated that the emerging core categories comprehensively accounted for the patterns observed across the data. The resulting selective coding structures provided a robust foundation for the next stage of the study: using the framework to compare open-source and closed-source LLMs in terms of how they approach energy-efficiency and environmental impacts.

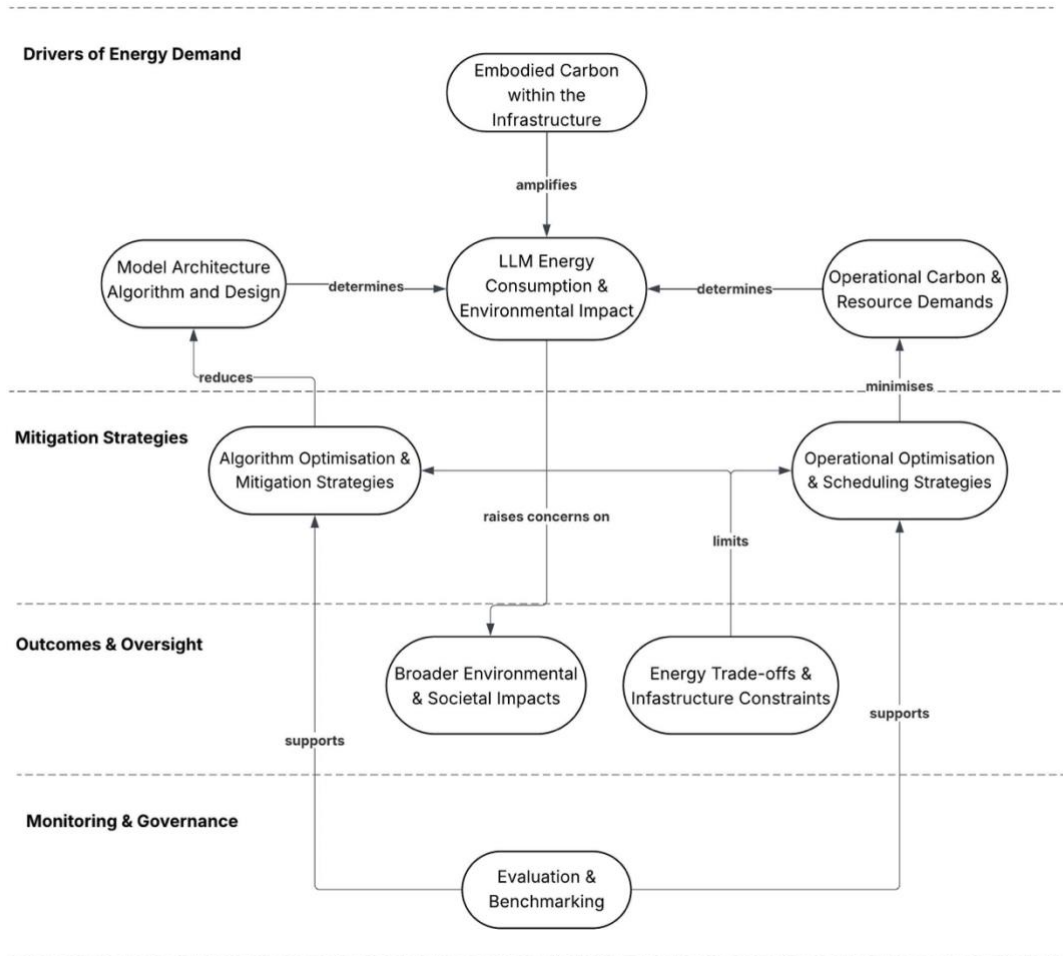


Figure 2: LLM Energy Consumption & Environmental Impact Framework

### 3.6 Ensuring Quality and Rigour

Several strategies were employed to ensure the study's quality and rigour:

- **Transparency:** An audit trail of the research process was maintained. All data sources, coding steps, and analytic memos were documented in Excel to make the process transparent and traceable.
- **Validation:** Quotes generated were regularly checked against the original data for validation.
- **Open Code Scope:** The relevance of the open codes was checked by the researcher against the quotes to ensure that they were coded correctly.

## 4. Findings

### 4.1 Findings Introduction

This section presents the core findings derived from the grounded theory analysis, structured around a conceptual framework developed to explain the drivers, mitigation strategies, outcome & oversight and governance influencing the energy demand of LLMs. Each subsection outlines a key thematic domain, beginning with the material and architectural sources of energy use, through which the study maps the interplay between infrastructure, mitigation strategies, outcomes and governance in shaping the environmental footprint of LLMs. The quotes referenced in this section can be found in Appendix 4: Findings Open Codes and Quotes.

### 4.2 Drivers of Energy Demand

The energy footprint of LLMs begins at the hardware level, where the scale, type, and lifecycle of computational infrastructure directly shape energy demand. This section explores the different drivers of energy-demand for LLMs, see Figure 3.

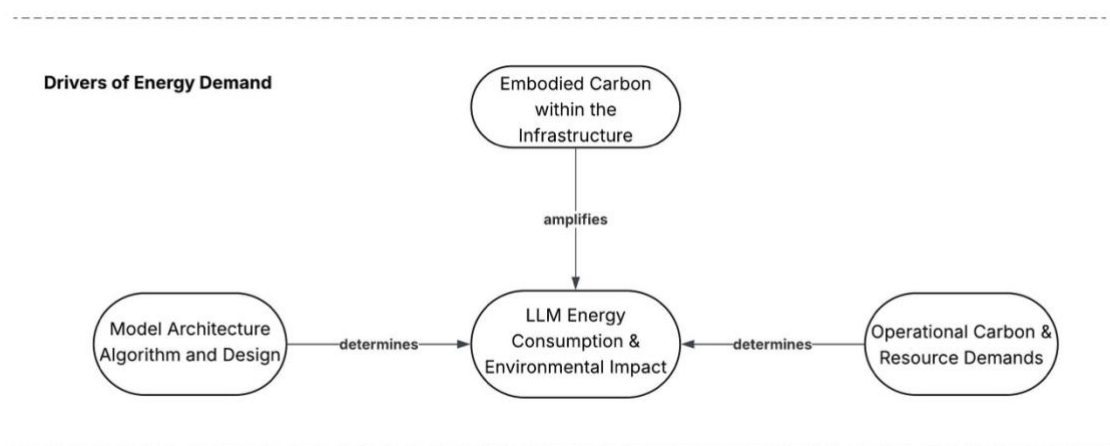


Figure 3: Drivers of Energy Demand

#### 4.2.1 Embodied Carbon within the Infrastructure

The hardware used for LLM development and deployment is directly linked to energy consumption. This massive energy demand stems from the hardware scale: cloud AI data

centres often host hundreds of thousands of processors with extensive cooling and power infrastructure. Beyond raw energy use, the embodied carbon associated with this hardware turnover presents a growing concern. As “Embodied Carbon from Hardware Turnover” highlights, see Quote 1.1. This underscores that to ensure sustainability the entire hardware lifecycle must be considered, not just runtime efficiency and the constant growing demand for computer machinery is concerning. As, this demand increases the hardware requirements for these algorithms increases and new data centres are built to meet this demand. This is a significant contribution to the embodied carbon.

Differences between open and closed-source LLMs is severe. Open models like LLaMA and DeepSeek favour modular and efficient deployment, often on personal devices or modest clusters. Closed-source models like GPT-4 and Claude, in contrast, rely on opaque, large-scale hardware deployments, which is highlighted by the theme “Reliance on Cutting-Edge Hardware”. Limited external evaluation and heightened “Embodied Carbon Emissions within the Device” (see Quote 1.3) due to centralised provisioning with less public observation are also concerns regarding closed-source models. It should also be noted that open-source models can be deployed in the cloud which would enhance their energy use. These emerging themes have identified a direct relationship between “Embodied Carbon within the Infrastructure” and the energy-demand of LLMs where LLMs deployed in bigger-scale or in the cloud utilise better hardware and therefore have lower energy-efficiency due to the embodied carbon of the infrastructure. Open-source models are more environmentally friendly in the sense that they provide alternative, more energy-efficient, architecture to be used for deployment such as smartphones. They don’t always require additional investments in hardware and are more flexible in their approach.

#### 4.2.2 Model Architecture and Algorithm Design

The architecture and design of an LLM have profound effects on its energy consumption, meaning model design choices may or may not amplify environmental impact. In general, larger models (with more parameters or layers) require more floating-point operations per inference or training step, thus consuming more energy. Argerich & Patino-Martinez (2024) empirically demonstrate that model size correlates strongly with power draw: increasing an LLM’s parameters and layers leads to higher energy use per query, all else equal. This is a

critical factor especially for larger systems. “Larger Model’s Energy Demand” reveals (referring to the inference stage), see Quote 2.1.

This might cause inefficiencies with tasks if the right model size is not selected carefully. Open-source LLMs tend to pursue leaner or more innovative architectures (smaller models with clever training or MoE designs) explicitly aimed at reducing computation, whereas closed LLMs often push model size and complexity to the extreme, relying on vast compute to maximise performance, see “Call for Leaner Inference Design”, Quote 2.2. This energy-consumption is further advanced by the energy-cost of training such large models on large data sets. This is also due to the incentive structure regarding open-source and closed-source LLMs where open-source LLMs are subject to criticism for their design and use of energy by research conducted by peers, which will be discussed later in this research. Another issue that is highlighted is the memory fragmentations and inefficiencies caused because of this. This is highlighted by the code “Unpredictable Output Lengths”, see Quote 2.3.

This quote highlights a major issue in memory management strategies of LLMs and this issue further impacts the energy-demand by preventing efficient use of hardware by the model and thus increasing the energy consumed for the tasks (Kwon, et al., 2023).

#### 4.2.3 Operational Carbon & Resource Demands

LLMs not only require powerful chips but also place heavy demands on digital infrastructure, the surrounding data centre environment needed to run those chips at scale. “LLM Energy Consumption & Environmental Impact” is partly determined by the efficiency and capacity of the operations while running this infrastructure. Modern AI data centres consist of dense racks of servers, specialised cooling systems (air and liquid cooling), and power delivery networks (McQuate, 2023).

Training state-of-the-art LLMs requires thousands of high-performance GPUs or TPUs running in parallel, each drawing significant power and generating heat that necessitates additional cooling (Mytton & Ashtine, 2022). While newer hardware is improving in efficiency, the availability of more powerful chips tends to amplify LLM energy use by enabling even larger models (Samsi, et al., 2023).

Running a large LLM service like ChatGPT means continuously operating and improving these infrastructures. The cost of this hardware (data centres) is highlighted by the “High per-Query Carbon Cost” theme which highlights, see Quote 3.1.

This further highlight that the infrastructure design of these facilities is another major environmental concern. A major part of the infrastructure footprint comes from cooling and electricity provisioning and the choice of the energy source as highlighted by the theme “Infrastructural Energy Source Dependency”, see Quote 3.2.

Additionally, data centres housing LLMs often draw on municipal power grids and must dissipate large amounts of heat. This significantly increases the energy-demand for computational tasks. While open-source LLMs provide the user with the freedom of deploying the model on their personal devices or in infrastructure with a smaller number of servers, the same effect applies when they are deployed in the cloud which is also a common practice for open-source LLMs since it provides users with additional benefits such as access to better hardware. These operational effects of the infrastructure are further enhanced by the high energy requirements of the inference stage of LLMs, highlighted by “Serving Phase Energy Dominance”, see Quote 3.3. These themes draw a direct correlation between the “Operational Carbon and Broader Resource Demands” and the Energy Consumption and Environmental Impact of LLMs.

## 4.3 Mitigation Strategies

As the environmental impact of LLMs becomes increasingly apparent, researchers have proposed a range of mitigation strategies aimed at reducing energy demand without compromising model functionality. This section explores two key pathways: carbon-focused algorithm optimisation and operational efficiency. These strategies work at different layers of the system to minimise emissions through smarter design and execution and therefore different applicability for open-source and closed-source LLMs.

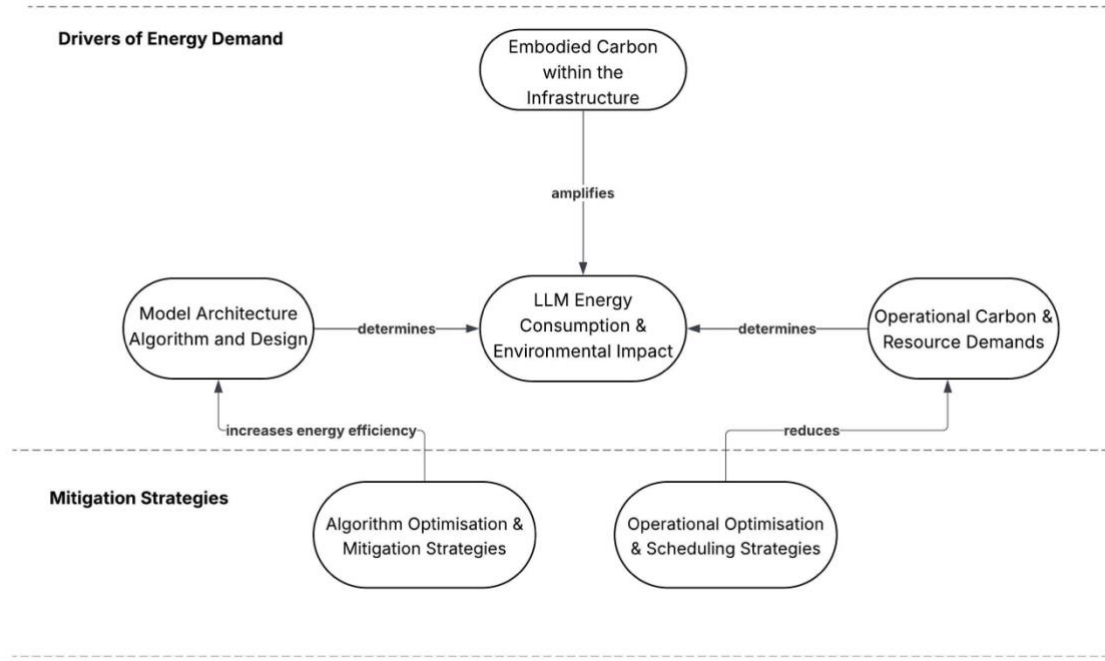


Figure 4: Energy Demand Mitigation Strategies

### 4.3.1 Algorithm Optimisation & Mitigation Strategies

Algorithm optimisation refers to strategies that explicitly aim to minimise the carbon footprint of LLMs through architectural and algorithmic modifications, rather than operations in general. Such strategies acknowledge that the structure and design of computation have a direct impact on the carbon intensity of the energy consumed. For instance, the code “Automated Sustainability Monitoring and Implementation” proposes, see Quote 4.1.

This directly aligns with the aim of algorithm-level mitigation. This represents a paradigm shift from traditional energy efficiency to carbon-awareness embedded within model output routines. Architectural techniques such as quantisation also play a major role. “Carbon-Driven Energy Saving” and “Quantisation-Based Carbon Reduction” highlight the use of reduced-precision computation to lower both FLOPs (floating-point operations) and resulting emissions, see Quote 4.2 and Quote 4.3. These are concrete examples of carbon-centric algorithmic reform.

Furthermore, “Carbon Efficient Generation Control” and “Algorithm Efficiency Imperative” (see Quote 4.4 and Quote 4.5) reflect deeper theoretical strategies: encoding carbon cost directly into optimisation objectives and computation constraints. This level of design

suggests a future where AI development is guided not only by performance or accuracy but also by embedded carbon limits. In essence, carbon-focused optimisation marks a growing emphasis on proactive carbon-aware design principles in algorithm development (Huang, et al., 2024). It reflects a maturing understanding that carbon efficiency is not just a hardware or infrastructure issue but can be meaningfully addressed at the level of mathematical models and software behaviours hence forming a direct relationship between the “Algorithm Optimisation & Mitigation Strategies ” and the “Model Architecture and Algorithm Design” categories.

#### 4.3.2 Operational Efficiency and Energy Sensitive Scheduling

“Operational Efficiency and Energy-Sensitive Scheduling” highlights mitigation strategies particularly suited to large-scale data centres, where high volumes of LLM workloads are processed continuously. These approaches focus on how tasks are scheduled, prioritised, and executed across hardware systems to maximise throughput while minimising energy use and emissions. Scheduling LLM workloads intelligently can cut emissions significantly, especially in large distributed systems where GPU utilisation and power scaling can be dynamically controlled. These highlight mitigation strategies that can be utilised in large data centres with bigger workloads. Solutions such as scheduling LLM workloads intelligently can cut emissions in larger systems. Whilst these methods can be effective for smaller systems, the impact is larger in larger-scale systems (Patterson, et al., 2021).

The overall finding is that “Operational Efficiency and Energy-Sensitive Scheduling Strategies” can minimise the emissions impact of LLMs without necessarily reducing their utility. For instance, the theme “Lack of Efficient Scheduling” proposes, see Quote 5.1. This reveals that when timing and workload distribution align with energy-efficient conditions, substantial emissions savings can be achieved without altering the model itself such as “Energy-Saving Intervention”, see Quote 5.2. Ultimately, these approaches demonstrate that energy demand is not only a function of what is computed, but also how and when it is computed (Patel, et al., 2024). Operational strategies thus complement algorithmic ones: together, they ensure that emissions are minimised without sacrificing the utility or responsiveness of LLM systems. This draws a direct connection between the “Operational Carbon and Resource Demands” and “Operational Optimisation and Scheduling Strategies”

providing a mitigation strategy where the infrastructure operations can be run more efficiently.

## 4.4 Outcomes & Oversight

As large language models continue to scale, their energy consumption generates outcomes that extend beyond technical boundaries, raising pressing environmental, ethical, and societal questions. This section explores how these impacts demand institutional oversight and thoughtful governance, linking energy demand to broader concerns about sustainability, infrastructure trade-offs, and the long-term social implications of AI deployment.

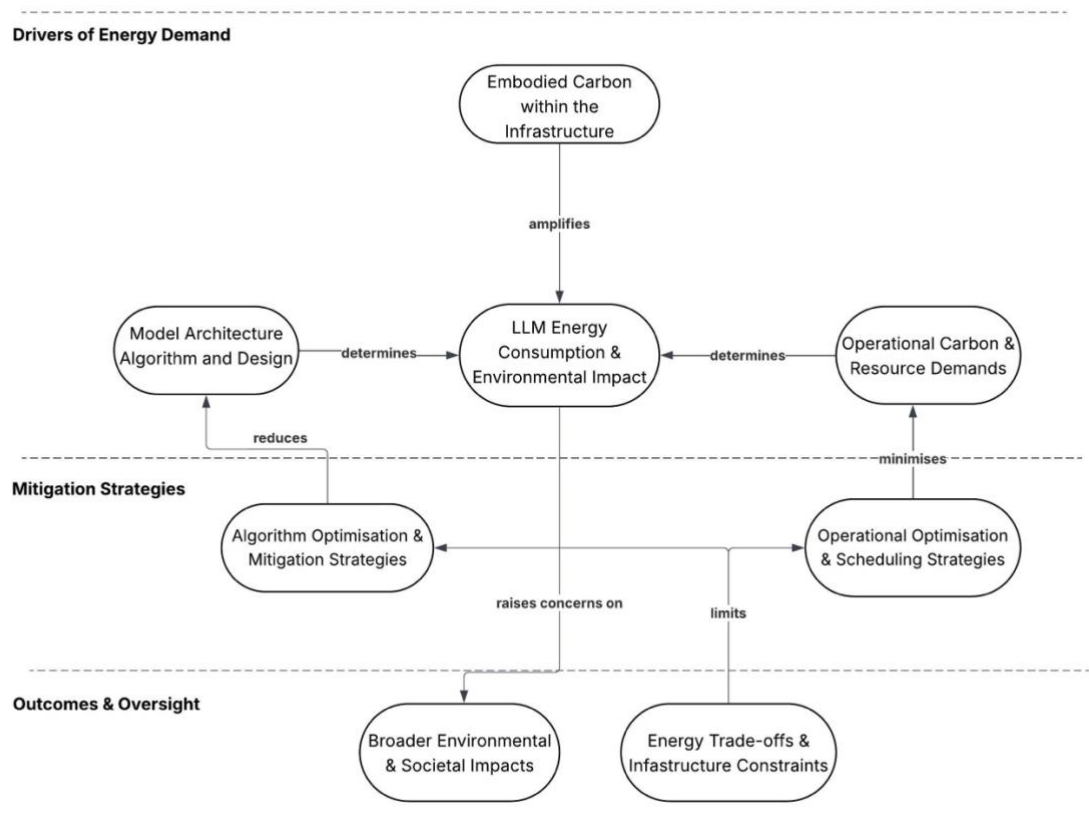


Figure 5: Outcomes & Oversight

### 4.4.1 Broader Environmental & Societal Impact

The increasing energy demand of LLMs has far-reaching consequences beyond the technical domain, extending into social, ethical, and ecological realms. “Environmental Cost of ML”

reflects the foundational concern that (regarding climate change), see Quote 6.1. Despite these concerns, some studies acknowledge potential upsides. “LLMs as Sustainability Enablers” and “Decarbonising Creative Work” suggest that LLMs could reduce the environmental footprint of certain activities, such as writing or coding, by offering energy-efficient alternatives to resource-intensive processes although it should be noted that this study considers simpler tasks, see Quote 6.2 and Quote 6.3 (Ren, et al., 2024). However, such optimism must be tempered with awareness “Balanced Deployment Ethics” calls for careful evaluation of when and how LLMs are deployed.

This tension underscores a broader societal challenge. “Socio-Technical Sustainability Challenge” and “Temporal Complexity of Impacts”(see Quote 6.4 and Quote 6.5) recognise that the effects of LLMs unfold over time, often in unpredictable ways, necessitating robust foresight. As such, institutional oversight becomes crucial. In sum, this category captures the pressing need to embed environmental ethics, social equity, and institutional transparency into the development trajectory of LLMs, ensuring that AI does not accelerate environmental degradation. This draws a link from the core category to the “Broader Environmental & Societal Impact” where the increasing energy-demand and environmental concerns increase this impact and level of societal concern.

#### 4.4.2 Energy Trade-offs and Infrastructure Constraints

The infrastructure supporting LLMs introduces substantial complexities when evaluating energy use and carbon emissions. A core challenge stems from the difficulty of accurately measuring carbon impact, as highlighted in the code “Challenges in Carbon Accounting”, which notes, see Quote 7.1.

These models operate across different energy grids, cooling systems, and geographic regions, each with varying carbon intensities, making it hard to assign a precise footprint to any given LLM task. Efforts to reduce these costs often lead to trade-offs with performance.

“Performance vs. Sustainability Trade-Off” reveals, see Quote 7.2. This reference articulates how increased performance through hardware increases the embodied carbon. Thus, energy optimisation must be carefully balanced against the performance of the hardware. These themes draw a connection where the “Energy Trade-offs and Infrastructure Constraints”

limits the “Embodied Carbon within the Infrastructure”. Further on this “Accuracy vs. Energy Demand Trade-off” highlights, see Quote 7.3.

It is important to prioritise energy savings when considering only minor improvements in accuracy. While open-source models have many tools available for algorithm-level optimisation, these tools often include trade-offs in performance, accuracy, quality of output... These trade-offs should be considered on a scenario basis, deciding on the priority of these metrics.

## 4.5 Monitoring & Governance

As the energy and environmental impact of LLMs comes under increasing inspection, the need for robust evaluation mechanisms has become more urgent. This section explores how emerging benchmarking frameworks aim to measure not only model performance but also carbon efficiency, and the challenges posed by the lack of transparency in closed-source systems.

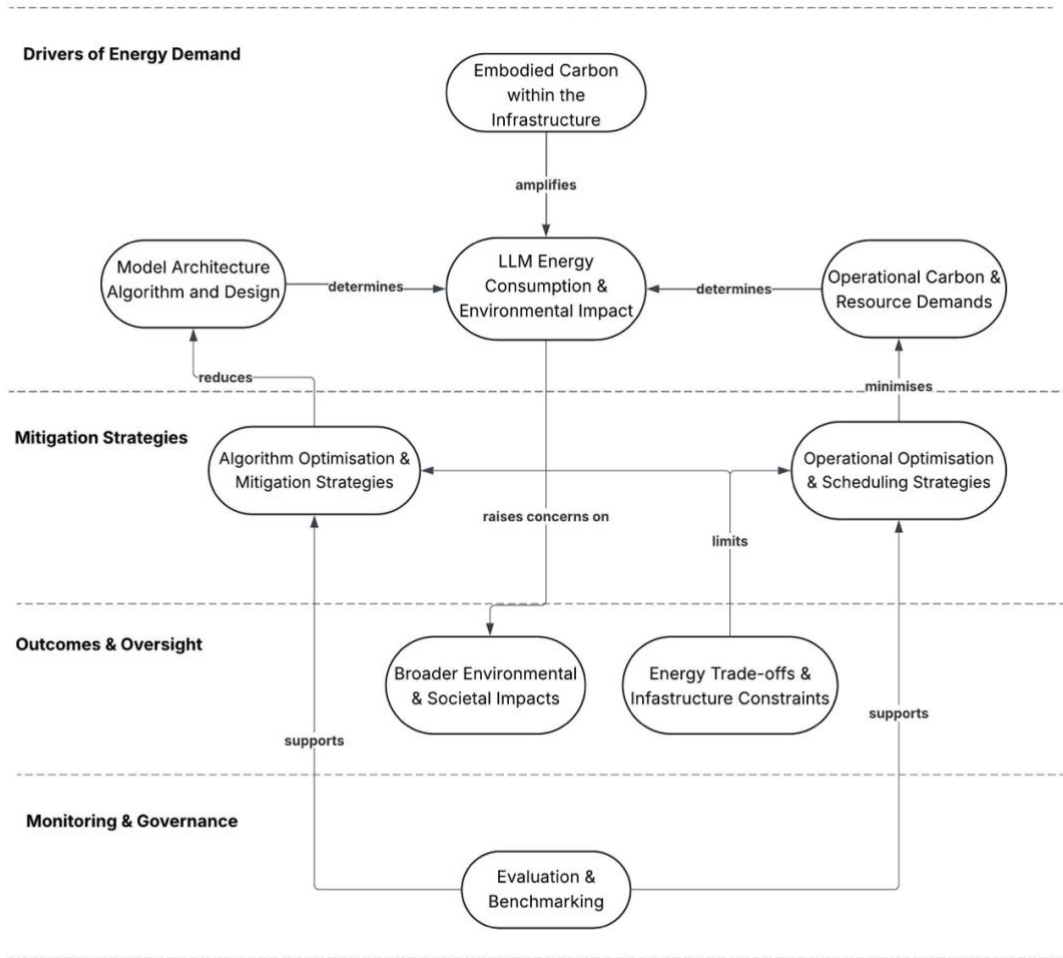


Figure 6: LLM Energy Consumption & Environmental Impact Framework

#### 4.5.1 Evaluation & Benchmarking

Effective evaluation and benchmarking are vital for promoting transparency and sustainability in LLM development. Yet, no standardised methodology currently exists across the industry, particularly for closed-source models. As “Unified Sustainability Benchmarking” states, see Quote 8.1.

This highlights the need for multi-dimensional unified assessments that account for both environmental cost and model performance. Most benchmarking in this field is driven by independent researchers evaluating open-source models, such as through tools like LoRA (Samsi, et al., 2023). In contrast, closed models like GPT-4 or Claude lack transparent frameworks, leaving emissions unreported and trade-offs unknown. Several frameworks attempt to bring consistency. “Functional Unit Standardisation” introduces, see Quote 8.2.

In summary, evaluation is evolving toward carbon-aware, context-sensitive tools, but the lack of adoption in closed-source AI remains a barrier to accountability and sustainable progress. This means that the “Evaluation & Benchmarking” supports mitigation strategies through transparency and accountability, specifically for open-source models.

## 5. Discussion

This discussion critically analyses how the design and deployment models of open-source and closed-source LLMs shape their energy demand, environmental impacts, and ability to mitigate these effects. Open-source LLMs like LLaMA and DeepSeek have demonstrated that it is possible to achieve high performance with comparatively limited or efficient use of hardware resources (DeepSeek, 2025) (Touvron, et al., 2023). This is mostly because open-source models such as LLaMA have models with different number of parameters that can be utilised for various tasks and DeepSeek's use of the MoE model, where the parameters are divided among many expert subnetworks, and for each given input token only a subset of those (e.g. one or two experts) are activated (Puigcerver, et al., 2024). This allows for more efficient use of hardware as open-source models can be deployed on local computers instead of high-performance data centres that have higher energy demands due to the additional costs through network traffic, data centre maintenance (cooling infrastructure etc.) and manufacturing (Mytton & Ashtine, 2022). Closed-source LLMs like GPT-4 and Claude, by contrast, are associated with massive, large-scale hardware deployments, often disclosed only through third-party estimates or indirect reports (Jiang, et al., 2024).

Research shows that not all additional parameters are equal in energy cost. Sparse or efficient architectures can maintain high performance while using less compute. Patterson et al. (2021) notes that large yet sparsely activated deep neural networks can achieve equivalent accuracy to large dense networks while consuming less than a tenth of the energy. This refers to techniques like MoE, where only a subset of model parameters is active for any given input. Closed-source architectures like those of GPT-4 and Claude are less transparent, but available information suggests they prioritise raw performance, potentially at the cost of higher energy use per model (Manchanda, et al., 2025).

Closed-source LLMs like GPT-4 and Claude rely heavily on large-scale, centralised infrastructure, which provides optimisation opportunities but they provide significant environmental impacts such as high-water usage and electricity demands. Conversely, open-source models like Meta's LLaMA and DeepSeek are typically utilised using smaller, decentralised hardware setups or shared public clusters. This approach reduces individual environmental footprints and makes hardware usage more flexible but may introduce redundant energy expenditures due to duplicated efforts across multiple groups.

Algorithmic optimisation offers notable energy-saving potential. Open-source communities actively innovate through techniques like quantisation and ‘generation directives’ to reduce energy consumption without compromising performance (Li, et al., 2025). These methods highlight a conscious, community-driven pursuit of carbon efficiency, reflecting Green AI principles. Closed-source providers similarly optimise models for efficiency but lack transparency, limiting external evaluation and broader adoption of effective practices.

Operational efficiency and energy-sensitive scheduling further distinguish these models. Closed-source providers can leverage centralised control, strategically shifting workloads and batch optimisation to reduce emissions (Patel, et al., 2024). In contrast, open-source deployments, often decentralised, face coordination challenges, making systematic scheduling difficult but still achievable through community efforts and shared tools.

Infrastructure constraints and energy trade-offs illustrate critical differences. Closed-source models frequently pursue maximum performance at high energy and resource costs, leading to practical limits as data centres face physical power and cooling constraints. Open-source models, however, naturally adapt to smaller-scale, efficient operation, with users selecting models suited to available hardware and efficiency requirements.

Finally, evaluation metrics and benchmarking for sustainability highlight an urgent need for standardised reporting of environmental impacts (Kneese & Young, 2024). Open-source models lead in transparency by voluntarily disclosing energy use and emissions, whereas closed-source models remain largely opaque. Practical recommendations include developing industry-wide sustainability benchmarks, mandatory carbon emission reporting in research publications, and researcher evaluation of closed-source models. The European Parliament has released the “Artificial Intelligence Act” where providers must document AI systems’ energy consumption throughout development and expected use, with the European Commission establishing standardised guidelines for clear, comparable, and verifiable reporting (European Parliament, 2023). These and similar guidelines should be employed globally.

In conclusion, both open and closed-source LLMs can mitigate environmental impacts through deliberate design, efficient scheduling, algorithmic innovation, and transparent

reporting. Bridging transparency gaps and collaboratively adopting best practices is crucial for sustainably harnessing the full potential of these powerful technologies. This research provided a framework to assess the energy-demand of LLMs.

## 6. Conclusion and Limitations

This study set out to explore how the design and deployment models of open-source and closed-source LLMs influence their energy demands, environmental impacts, and their capacities for mitigation. Using a qualitative research methodology guided by grounded theory, the study provided comprehensive insights into the factors that drive energy demand, the mitigation strategies employed, and the governance mechanisms influencing these processes.

Findings from this study indicate that open-source models such as Meta's LLaMA and DeepSeek offer significant advantages in terms of energy efficiency through their modular designs and computational optimisation strategies. Their accessible and transparent architecture encourages innovation in efficiency-focused solutions, while also distributing computational demands, thus mitigating concentrated environmental impacts typically associated with large-scale data centres (DeepSeek, 2025).

Conversely, closed-source models like OpenAI's GPT series and Anthropic's Claude demonstrate higher energy consumption and environmental footprints due to their centralised infrastructures and opaque reporting standards. The concentration of computational power required for these closed-source models not only results in substantial energy and water use but also creates barriers for external accountability and sustainability assessment.

The limitations of this research include the relatively limited availability of existing literature due to the novel nature of the topic, potentially restricting the depth and breadth of insights. Additionally, there is greater availability of research data on open-source models compared to closed-source models, which may introduce biases in comparative assessments. Furthermore, the lack of transparency in the strategies and practices employed by closed-source LLM providers hampers comprehensive evaluations and limits the ability to validate claims regarding their sustainability measures. The use of ChatGPT also poses ethical challenges, such as worries about data privacy, the danger of fostering echo chambers that isolate users from diverse viewpoints, and the potential for the model to produce false or misleading content—so-called “hallucinations” (Alkaissi & McFarlane, 2023).

In conclusion, this research underscores the importance of transparent governance structures in managing the significant environmental impacts associated with LLMs, specifically because of the vast number of factors affecting the energy-efficiency and the bespoke nature of the algorithms. Open-source models present a more sustainable path forward, given their inherent transparency and greater community engagement, which facilitate ongoing improvements in energy efficiency and mitigation strategies. However, closed-source models can leverage centralised control to swiftly implement significant efficiency upgrades, provided there is an increase in transparency and accountability. The unique nature of all algorithms also highlights the need for higher transparency, specifically for researchers.

## 7. Implications for Future Research

To consolidate the nascent field of sustainable LLM research, future scholarship should prioritise three complementary directions:

1. Operational transparency for closed-source LLMs

Comparative work is currently hamstrung by the opacity of proprietary providers; systematic disclosure of energy, water and hardware metrics is rare. Studies should design auditable reporting protocols for closed-source LLMs.

2. Standardised, multi-dimensional benchmarking

The field lacks harmonised metrics that integrate functional-unit energy use with life-cycle carbon and water footprints. Extending emerging frameworks such as FUEL (Wu, et al., 2025) and unified sustainability benchmarking proposals (Ding & Shi, 2024), and aligning them with forthcoming EU disclosure duties under the Artificial Intelligence Act, should be a priority.

3. Governance and incentive structures.

Future work should interrogate how carbon pricing, green procurement and mandatory reporting influence organisational behaviour, and how governance models can embed energy justice concerns. Comparative policy studies and participatory action research with developers, utilities and affected communities will illuminate pathways that align innovation with infinite boundaries.

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## 9. Appendix

### 9.1 Appendix 1: Structure of the Custom GPT Model

|                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Name                     | "QualitativeEdge"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Link                     | <a href="https://chatgpt.com/g/g-6810a54d18788191b7734648866feb16-qualitativeedge">https://chatgpt.com/g/g-6810a54d18788191b7734648866feb16-qualitativeedge</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Description              | An expert in qualitative data analysis, this GPT conducts a thematic analysis of documents by applying open coding practices.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Instructions             | <p>Accuracy and Relevance: Ensure that the text codes represent the researchers text content and themes. Prioritise information relating to your field.</p> <p>Scholarly Tone: Adopt a scholarly, academic tone using British English.</p> <p>Objectivity: Maintain objectivity throughout the whole process.</p> <p>Confirmation Bias: Avoid only focusing on information that supports your theory but approach the task critically.</p> <p>Language: Ensure your replies are in British English</p>                                                                                                                                                                                                                                                                                                                   |
| Knowledge Document Names | <ol style="list-style-type: none"> <li>1. A Step-by-Step Process of Thematic Analysis to Develop a Conceptual Model in Qualitative Research (Naeem, et al., 2023)</li> <li>2. Basics of Qualitative Research (3<sup>rd</sup> Ed): Analyzing Data for Concepts</li> <li>3. Thematic Analysis of qualitative data: AMEE Guide No. 131</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| User Input               | <p>"You are an expert qualitative thematic analysis researcher in the field of LLM energy-efficiency. You will assist in qualitative text analysis and coding data. Follow the instructions below step by step.</p> <ol style="list-style-type: none"> <li>1. Read the document provided thoroughly and comprehend the research undertaken.</li> <li>2. Identify key ideas, arguments, themes, and content in the text provided.</li> <li>3. Analyse the entirety of the text to identify significant, meaningful phrases for the research within the topic of energy-efficiency.</li> <li>4. Generate initial codes/labels for the significant text you have identified where each code represents a semantic unit corresponding to the text. Codes are concise labels that capture the key ideas, meanings,</li> </ol> |

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <p>or emotional resonance of passages in a text. By coding, you break large volumes of data into small, precise units, making the material easier to organise and analyse. The codes should be well-defined and should relate to the study undertaken. The codes should be general. You should identify and define each code in the text and maintain your objectivity throughout.</p> <p>5. Ensure each code is supported by a direct quote from the paper.</p> <p>6. Present your final codes in a table format with direct quotes and the related codes.”</p> |
|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## 9.2 Appendix 2: Journal Articles used for Qualitative Analysis

|         |                                                                                                    |
|---------|----------------------------------------------------------------------------------------------------|
| Paper 1 | <i>Estimating the Carbon Footprint of Bloom, a 176B Parameter Model</i>                            |
| Paper 2 | <i>Sustainable LLM Serving: Environmental Implications, Challenges, and Opportunities</i>          |
| Paper 3 | <i>Efficient Memory Management for Large Language Model Serving with PagedAttention</i>            |
| Paper 4 | <i>SPROUT: Green Generative AI with Carbon-Efficient LLM Inference</i>                             |
| Paper 5 | <i>Characterizing Power Management Opportunities for LLMs in the Cloud</i>                         |
| Paper 6 | <i>Unveiling Environmental Impacts of Large Language Model Serving: A Functional Unit View</i>     |
| Paper 7 | <i>Reconciling the contrasting narratives on the environmental impact of large language models</i> |
| Paper 8 | <i>Risks and Benefits of Large Language Models for the Environment</i>                             |
| Paper 9 | <i>Towards Green ai In Fine-Tuning Large Language Models via Adaptive Backpropagation</i>          |

|          |                                                                                             |
|----------|---------------------------------------------------------------------------------------------|
| Paper 10 | <i>From Words to Watts: Benchmarking the Energy Costs of Large Language Model Inference</i> |
|----------|---------------------------------------------------------------------------------------------|

### 9.3 Appendix 3: Sample Open Coding by the Custom GPT Model "QualitativeEdge"

| Quote                                                                                                                                                                                                                                                                                                                                           | Code                                           | Category                                                 |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|----------------------------------------------------------|
| <i>"Newer hardware offers better performance but is not always greener due to higher embodied carbon. Older hardware can lower carbon emissions while meeting quality and performance constraints."</i> (Wu, et al., 2025)                                                                                                                      | "Newer Hardware on Embodied Carbon"            | "Embodied Carbon within the Infrastructure"              |
| <i>"SPROUT introduces the concept of "generation directives", a strategy to indirectly manipulate the number of autoregressive inference iterations while providing high-quality content generation."</i> (Li, et al., 2025)                                                                                                                    | "Inference Carbon Reduction with Directives"   | "Algorithm Optimisation and Mitigation Strategies"       |
| <i>"simple interventions like GPU power capping is available to be deployed today - our preliminary analysis with LLM inference in this paper suggests that power capping can be an effective tool for reducing inference energy."</i> (Samsi, et al., 2023)                                                                                    | "Power Capping for Mitigating High Energy Use" | "Operational Efficiency and Energy-Sensitive Scheduling" |
| <i>"Advanced hardware offers higher performance but is not always greener due to higher embodied carbon"</i> (Wu, et al., 2025)                                                                                                                                                                                                                 | "Performance vs Sustainability Trade-Off"      | "Energy Trade-offs and Infrastructure Constraints"       |
| <i>"sustainability is not, at its root, a technical problem, nor will merely technical solutions be sufficient. Instead, deep economic, political, and cultural adjustments will ultimately be required, along with a major, long-term commitment in each sphere to deploy the requisite technical solutions at scale."</i> (Ren, et al., 2024) | "Socio-Technical Sustainability Challenge"     | "Broader Environmental and Societal Impacts"             |
| <i>"The AI boom, driven by the demand for generative artificial intelligence (GenAI) has prompted concerns over its</i>                                                                                                                                                                                                                         | "Carbon Emission Concerns"                     | "Operational Carbon & Resource Demands"                  |

|                                                                                                                                                                                                                                                                            |                                                |                                                          |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|----------------------------------------------------------|
| <i>environmental impact, particularly in terms of carbon emissions associated with the datacenters hosting these technologies.</i> " (Li, et al., 2025)                                                                                                                    | Regarding Data Centres"                        |                                                          |
| <i>"LLMs deployed in customer operated virtual machines prevent provider visibility into workloads."</i> (Patel, et al., 2024)                                                                                                                                             | "Barriers to Evaluation of Energy Consumption" | "Evaluation Metrics and Benchmarking for Sustainability" |
| <i>"Our measurement on a single Nvidia A100 GPU over 5,000 testing prompts without optimal batching showed that Gemma-2B-it consumes about 0.0002 kWh for producing a 500-word response, roughly 1% energy of Llama-3-70B for doing the same job."</i> (Ren, et al., 2024) | "Carbon-Impact and Size Correlation"           | "Model Architecture and Algorithm Design"                |

## 9.4 Appendix 4: Findings Open Codes and Quotes

| Quote Number | Axial Category                              | Open Code                                     | Quote                                                                                                                                                                                                                                  |
|--------------|---------------------------------------------|-----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Quote 1.1    | "Embodied Carbon within the Infrastructure" | "Embodied Carbon from Hardware Turnover"      | <i>"Heavy reliance on the latest GPUs to improve LLM serving performance significantly raises embodied carbon footprints."</i> (Ding & Shi, 2024)                                                                                      |
| Quote 1.2    | "Embodied Carbon within the Infrastructure" | "Reliance on Cutting-Edge Hardware"           | <i>"Existing performance optimization techniques for LLM serving highly rely on the latest and most advanced hardware, as older hardware often cannot meet the compute demands and latency SLOs of LLM models."</i> (Ding & Shi, 2024) |
| Quote 1.3    | "Embodied Carbon within the Infrastructure" | "Embodied Carbon Emissions within the Device" | <i>"Embodied carbon represents the carbon emissions associated with the manufacturing and packaging of computer components, effectively "embodied" within the device itself."</i> (Li, et al., 2025)                                   |
| Quote 2.1    | "Model Architecture and Algorithm Design"   | "Larger Model's Energy Demand"                | <i>"We first observe that under the same configuration, larger models (e.g., BLOOM- 176B) incur a much higher amount of computation during both prompt and token phases and show significantly larger peak and mean</i>                |

|           |                                           |                                            |                                                                                                                                                                                                                                                                                                                                                                            |
|-----------|-------------------------------------------|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|           |                                           |                                            | <i>power consumption.</i> ” (Patel, et al., 2024)                                                                                                                                                                                                                                                                                                                          |
| Quote 2.2 | “Model Architecture and Algorithm Design” | “Call for Leaner Inference Design”         | <i>“as AI compute requirements have increased, there is an increasing focus on approaches to reduce the carbon and energy footprints of datacenters by making larger models leaner or more efficient.”</i> (Samsi, et al., 2023)                                                                                                                                           |
| Quote 2.3 | “Model Architecture and Algorithm Design” | “Unpredictable Output Lengths”             | <i>“Due to the unpredictable output lengths from the LLM, they statically allocate a chunk of memory for a request based on the request’s maximum possible sequence length, irrespective of the actual input or eventual output length of the request.”</i> (Kwon, et al., 2023)                                                                                           |
| Quote 3.1 | “Operational Carbon & Resource Demands”   | “High Per-Query Carbon Cost”               | <i>“serving a single prompt in ChatGPT produces more than 4 grams of CO<sub>2</sub>eq, which is over 20 times the operational carbon footprint of a web search query.”</i> (Li, et al., 2025)                                                                                                                                                                              |
| Quote 3.2 | “Operational Carbon & Resource Demands”   | “Infrastructural Energy Source Dependency” | <i>“The carbon footprint will depend on the energy use and the carbon intensity of the energy source being used. In addition to carbon dioxide emissions, the computing facilities may also have other environmental impacts such as water use and soil pollution or sealing, which could have broader implications for environmental quality.”</i> (Rillig, et al., 2023) |
| Quote 3.3 | “Operational Carbon & Resource Demands”   | “Serving Phase Energy Dominance”           | <i>“Although training takes months and consumes significant amounts of energy, recent studies show that the energy consumption of LLM serving has now surpassed that of training,</i>                                                                                                                                                                                      |

|           |                                                          |                                                          |                                                                                                                                                                                                       |
|-----------|----------------------------------------------------------|----------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|           |                                                          |                                                          | leading to significant environmental impacts, especially in terms of carbon footprints." (Li, et al., 2025)                                                                                           |
| Quote 4.1 | "Algorithm Optimisation & Mitigation Strategies"         | "Automated Sustainability Monitoring and Implementation" | <i>"Design, implementation and evaluation of carbon-friendly generation directive configuration for LLM inference."</i> (Li, et al., 2025)                                                            |
| Quote 4.2 | "Algorithm Optimisation & Mitigation Strategies"         | "Carbon-Driven Energy Saving"                            | <i>"High carbon footprint, instead, leads to a higher objective of FLOPs reduction for better embracing Green AI."</i> (Huang, et al., 2024)                                                          |
| Quote 4.3 | "Algorithm Optimisation & Mitigation Strategies"         | "Quantisation-Based Carbon Reduction"                    | <i>"Quantization significantly lowers carbon emissions, especially for larger models"</i> (Wu, et al., 2025)                                                                                          |
| Quote 4.4 | "Algorithm Optimisation & Mitigation Strategies"         | "Carbon Efficient Generation Control"                    | <i>"SPROUT introduces "generation directives" to guide the autoregressive generation process, achieving a balance between ecological sustainability and high-quality outputs."</i> (Li, et al., 2025) |
| Quote 4.5 | "Algorithm Optimisation & Mitigation Strategies"         | "Algorithm Efficiency Imperative"                        | <i>"With both the training of LLMs and inference having large energy demands, prompting an early call for algorithmic efficiency"</i> (Rillig, et al., 2023)                                          |
| Quote 5.1 | "Operational Efficiency and Energy Sensitive Scheduling" | "Lack of Efficient Scheduling"                           | <i>"This lack of correlation is due to the variation in arrival times and scheduling at cluster scale."</i> (Patel, et al., 2024)                                                                     |
| Quote 5.2 | "Operational Efficiency and Energy Sensitive Scheduling" | "Energy-Saving Intervention"                             | <i>"power capping as an energy savings intervention can be effective when applied appropriately."</i> (Samsi, et al., 2023)                                                                           |
| Quote 6.1 | "Broader Environmental & Societal Impact"                | "Environmental Cost of ML"                               | <i>"estimating and reducing our carbon emissions is an important part of mitigating its impacts."</i> (Luccioni, et al., 2023)                                                                        |
| Quote 6.2 | "Broader Environmental & Societal Impact"                | "LLMs as Sustainability Enablers"                        | <i>"LLMs can lead to more sustainable alternatives to current practices"</i> (Ren, et al., 2024)                                                                                                      |

|           |                                                    |                                            |                                                                                                                                                                                                                                                                                                                                                 |
|-----------|----------------------------------------------------|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Quote 6.3 | “Broader Environmental & Societal Impact”          | “Decarbonising Creative Work”              | <i>“These findings emphasize the potential of LLMs to reduce the environmental impact of knowledge work and creative tasks, while simultaneously reducing costs.” (Ren, et al., 2024)</i>                                                                                                                                                       |
| Quote 6.4 | “Broader Environmental & Societal Impact”          | “Socio-Technical Sustainability Challenge  | <i>“sustainability is not, at its root, a technical problem, nor will merely technical solutions be sufficient. Instead, deep economic, political, and cultural adjustments will ultimately be required, along with a major, long-term commitment in each sphere to deploy the requisite technical solutions at scale.” (Ren, et al., 2024)</i> |
| Quote 6.5 | “Broader Environmental & Societal Impact”          | “Temporal Complexity of Impacts”           | <i>“LLM adoption will likely have immediate, mid-term, and long-term consequences for the economy and society.” (Ren, et al., 2024)</i>                                                                                                                                                                                                         |
| Quote 7.1 | “Energy Trade-offs and Infrastructure Constraints” | “Challenges in Carbon Accounting”          | <i>“the difficulty of precisely estimating the carbon footprint of ML models” (Luccioni, et al., 2023)</i>                                                                                                                                                                                                                                      |
| Quote 7.2 | “Energy Trade-offs and Infrastructure Constraints” | “Performance vs. Sustainability Trade-Off” | <i>“Advanced hardware offers higher performance but is not always greener due to higher embodied carbon” (Wu, et al., 2025)</i>                                                                                                                                                                                                                 |
| Quote 7.3 | “Energy Trade-offs and Infrastructure Constraints” | “Accuracy vs. Energy Demand Trade-off”     | <i>“Performing more frequent tensor importance evaluation within each epoch brings little improvement on the task accuracy but noticeably increase the training cost.” (Huang, et al., 2024)</i>                                                                                                                                                |
| Quote 8.1 | “Evaluation & Benchmarking”                        | “Unified Sustainability Benchmarking”      | <i>“to build a generalized benchmarking framework that bridges different LLM optimization techniques with various models and libraries, facilitating seamless integration and evaluation” (Ding &amp; Shi, 2024)</i>                                                                                                                            |

|           |                             |                                   |                                                                                                                                                                                            |
|-----------|-----------------------------|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Quote 8.2 | “Evaluation & Benchmarking” | “Functional Unit Standardisation” | <i>“We introduce FUEL, the first evaluation framework for unveiling LLM serving’s environmental impact by leveraging functional units as the basis for comparison.” (Wu, et al., 2025)</i> |
|-----------|-----------------------------|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|